

Motivation



Figures 1 and 2 illustrate QR code scanning issues generated using Gemini 3 Pro.

- ❖ The figure highlights real-world conditions such as motion blur, geometric distortion, environmental effects, and poor print quality that frequently cause **QR scanning failures**, emphasizing the limited reliability of QR codes in practical use.

Proposed Dataset

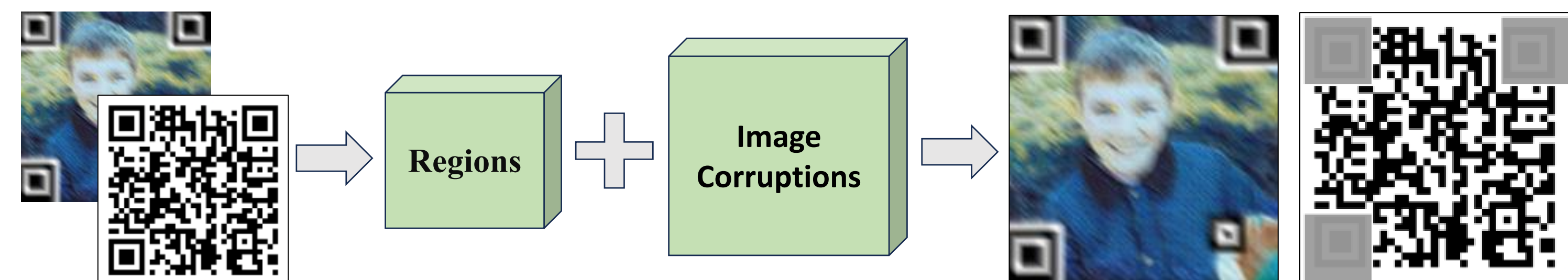


Figure 3: The pipeline how we generate the corrupted QR code in global and local.

- ❖ We construct a comprehensive dataset of corrupted QR codes (standard and aesthetic) by applying 18 real- world image corruption techniques [1] in two different regions Global and local.

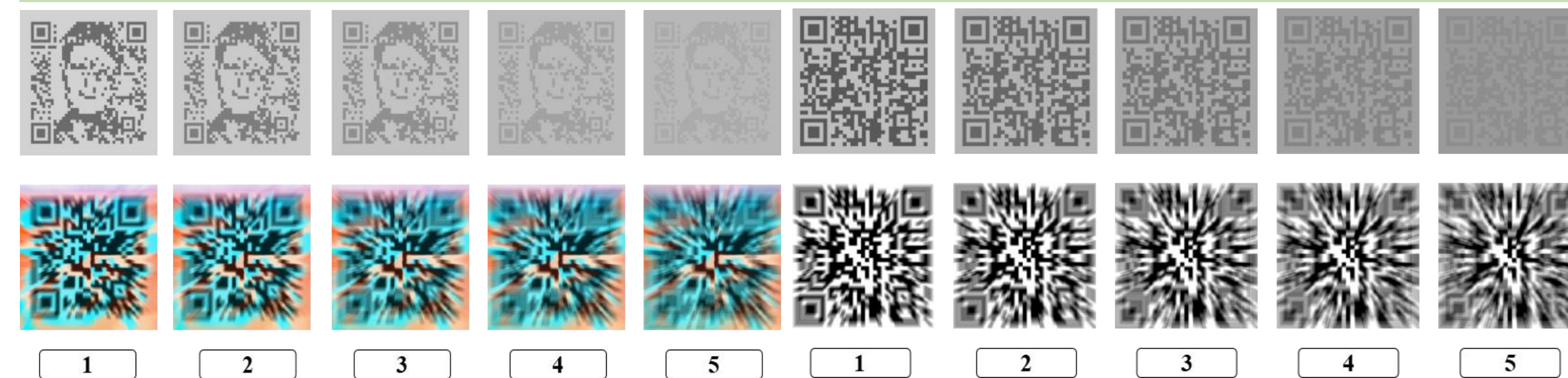
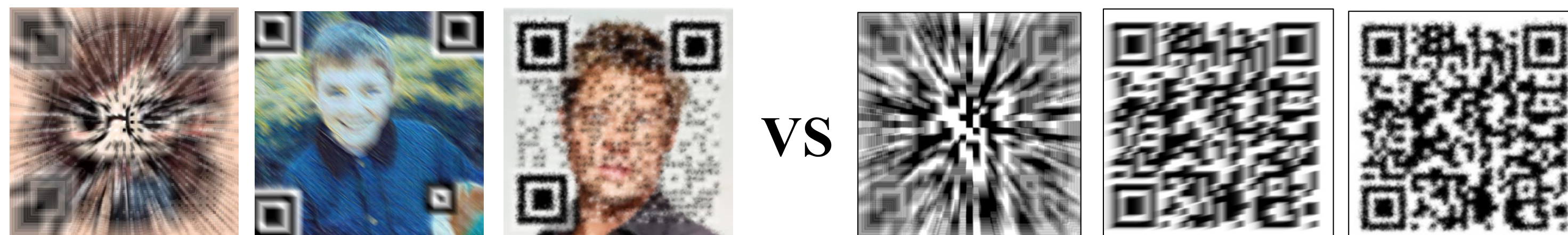


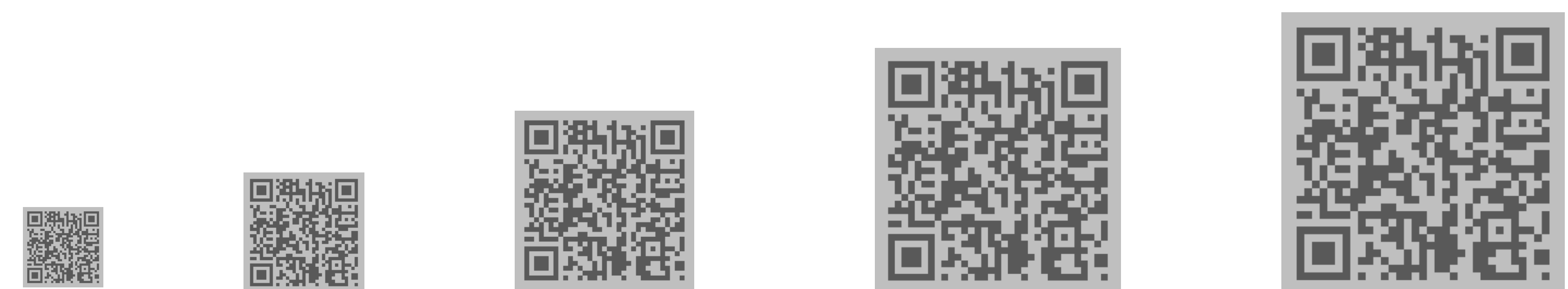
Figure 4: Examples of global QR corruptions at severity levels 1–5.

- ❖ We categorize 18 distinct corruptions into 5 major real-world scenarios: Environmental Effects (EE), Motion/Focus Issues (MFI), Print/Display Imperfections (PDI), Lighting/Color Variation (LCV), and Physical Damage/Obstruction (PDO).

Key Findings



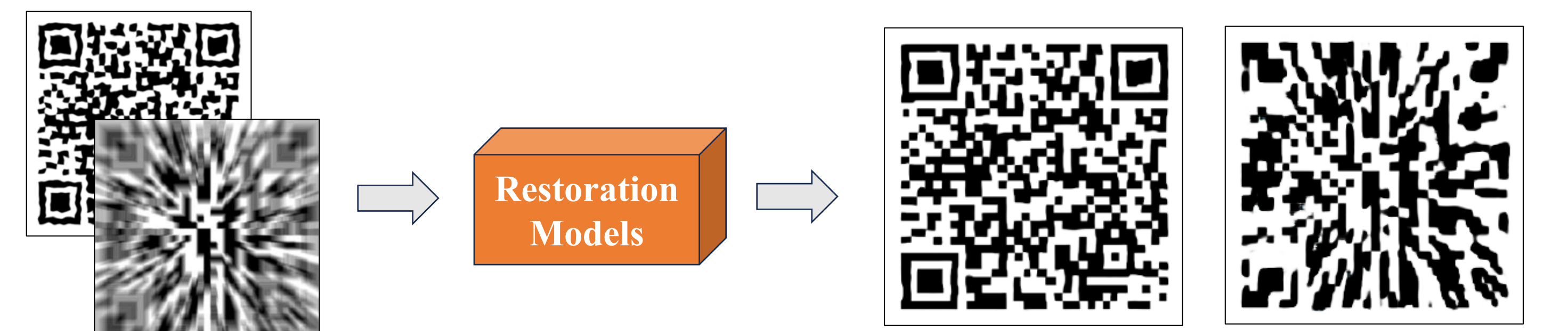
- ❖ Aesthetic QR code more vulnerable than standard QR code to image corruptions



- ❖ Small QR codes are more vulnerable to real-world image corruptions.



- ❖ Finder-region distortions are the most vulnerable, reduce decoding reliability.



- ❖ Existing models perform well on seen corruptions but fail to generalize to **unseen corruption types**.

References

- [1] Hendrycks, D.; and Dietterich, T. 2019. Benchmarking neural network robustness to common corruptions and perturbations. arXiv preprint arXiv:1903.12261.
- [2] Hao Dong, Haibin Liu, Mingfei Li, Fujie Ren, and Feng Xie. An algorithm for the recognition of motion-blurred qr codes based on generative adversarial networks and attention mechanisms. International Journal of Computational Intelligence Systems, 17(1):83, 2024.
- [3] Gu, Wencheng, et al. "Gs-DeblurGANv2: a QR code deblurring algorithm based on lightweight network structure." Multimedia Systems 30.2 (2024): 87.

Experimental Results and Analysis

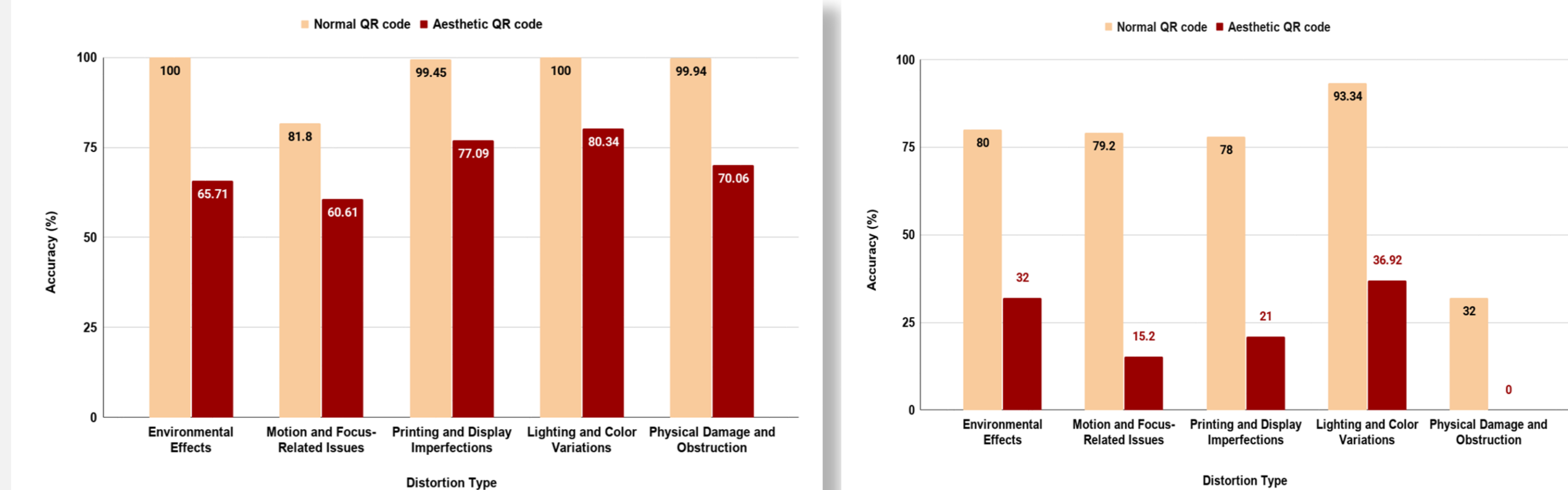


Figure 5. standard vs. aesthetic QR code scannability by QReader (left) and ZXing (right).

- ❖ **Aesthetic QR codes are more vulnerable to real-world distortions** than standard QR codes, with the strongest decoding degradation observed under motion- and focus-related distortions across both scanners.

Distortions	Pyzbar			OpenCV			QReader			ZXing			Google Lens			TeaCapps		
	F	W	B	F	W	B	F	W	B	F	W	B	F	W	B	F	W	B
EE	0	100	63.54	7.54	96.8	66.34	0	100	100	54	97.34	93.34	89.34	100	100	100	100	100
MFI	0	55.08	64.6	15.16	61.6	93.36	24.84	84.88	91.92	78.4	71.2	70.4	89.6	80	97.6	100	100	84
PDI	0.1	63.45	97.45	7.27	70.35	98.05	2.2	100	100	62	83	97	100	80	100	100	100	100
LCV	0	100	100	8.64	98.4	98.54	6.6	100	100	86.67	100	100	90.67	100	100	93.34	100	100
PDO	6.67	32.94	67.14	22.29	51.2	62.6	84.34	100	100	28	29.34	88	100	78.67	100	100	100	100

Table 1. Decoding success rate across regions. F = Finder, W = White, B = Black.

- ❖ The results shows that finder-region distortions are more harmful than global distortions, with MFI-related localized distortions causing severe decoding failures across scanners.

Model	Trained on	RR	PSNR	SSIM
No Defense (Motion)	--	69.70	14.53	0.63
No Defense (Zoom)	--	51.60	13.90	0.60
Dong et al. 2024 [2]	Motion Blur	79.79	16.55	0.76
Dong et al. 2024 [2]	Zoom Blur	77.32	22.80	0.91
Gu et al. 2024 [3]	Motion Blur	62.97	13.07	0.69
Gu et al. 2024 [3]	Zoom Blur	79.75	23.88	0.90

Tab2. Comparison of two GAN restoration models evaluated on unseen corruptions.

- ❖ Existing models provide only limited defense against unseen corruptions, with Dong et al. showing better generalization than Gu et al.,
- ❖ while diverse multi-corruption training substantially improves robustness, this limitation give future work on unseen generalization.